

Q1.

DIRECTIONS for questions 1 and 2: Select the correct alternative from the given choices.

After a batsman scored 35, 46 and 39 runs in his fifth, sixth and seventh innings, he observed that his average runs scored per inning in his first seven innings was less than that in his first four innings. If his average runs scored per inning in his first eight innings was equal to 39, then the number of runs that he scored in his eighth inning was at most

☐ a)

28.

☐ b)

31.

☐ c)

71.

☒ d)

68.

Q2. DIRECTIONS *for questions 1 and 2:* Select the correct alternative from the given choices.

The graph of a quadratic expression, $ax^2 + bx + c$, attains a maximum value of 10 at $x = 2$. If the graph intersects the x -axis at two points, one positive and the other negative, then which of a , b and c is/are positive?

- ☐ a) **Only c**
- ☐ b) **Only b**
- ☐ c) **Both b and c**
- ☐ d) **Both a and c**

Q3.

DIRECTIONS *for question 3:* Type in your answer in the input box provided below the question.

A group of workers in a puppet manufacturing company was assigned the task of making 395 puppets. Each worker in the group works at a constant rate and makes five puppets per hour. At the end of the first hour, some of the workers were reassigned to another task; at the end of the second hour, the same number of the remaining workers were also reassigned to another task, and a similar reassignment occurred at the end of every hour till the task was complete. If the entire task was completed in exactly 4 hours and 20 minutes, how many puppets were finished in the first three hours?

Q4. DIRECTIONS *for question 4:* Select the correct alternative from the given choices.

The number of positive integers from 1 to 10^6 (both inclusive), which are perfect squares, perfect cubes or perfect 4th powers is

☐ a) 1116.

☐ b) 1100.

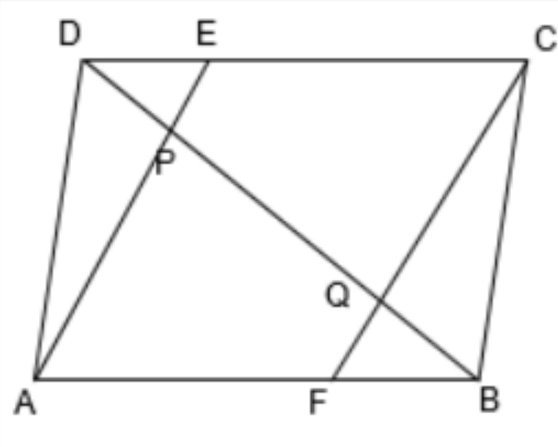
☐ c) 1110.

☐ d) 1090.

Q5. DIRECTIONS *for questions 5 and 6:* Type in your answer in the input box provided below the question.

The difference between the compound interest and the simple interest for two years on a certain sum at a certain rate of interest is ₹900. If the compound interest for the second year is ₹6900, find the principal (in ₹).

Q6. DIRECTIONS for questions 5 and 6: Type in your answer in the input box provided below the question.



In the figure above, ABCD is a parallelogram, where $DE : EC = BF : FA = 2 : 5$. If $BD = 18$ cm, what is the length (in cm) of PQ?

Q7. DIRECTIONS *for questions 7 and 8:* Select the correct alternative from the given choices.

Find the value of $\sqrt{11 - \sqrt{72}} + \sqrt{11 + \sqrt{72}}$.

☐ a)

$$2\sqrt{3}$$

☐ b)

$$4\sqrt{3}$$

☐ c) **6**

☐ d)

$$4\sqrt{2}$$

Q8. DIRECTIONS *for questions 7 and 8:* Select the correct alternative from the given choices.

If abc is a three-digit number in the number system to the base 8, how many such numbers satisfy both of the following conditions?

I. The number lies between $(101)_{10}$ and $(372)_{10}$.

II. $7 > b > c > 2$.

☐ a) 30

☐ b) 29

☐ c) 28

☐ d) 27

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

Find the value of $\sin^2 A + \sin^2 B + \sin^2 C$, if A, B and C are the angles of a right-angled triangle.

Q10. DIRECTIONS *for question 10:* Select the correct alternative from the given choices.

In a polygon, let X be the number of triangles that can be formed by joining any three of the vertices of the polygon, such that exactly one of the sides of the triangle is also a side of the polygon. Further, let Y be the number of triangles which can be formed in a similar manner, but with exactly two of the sides of the triangle also being the sides of the polygon. If, in a regular convex polygon, P , $X + Y = 28$, then P is

- ☐ a) a decagon.
- ☐ b) a nonagon.
- ☐ c) an octagon.
- ☐ d) a heptagon.

Q11. DIRECTIONS *for question 11:* Type in your answer in the input box provided below the question.

If K is a non-negative integer less than 200, such that there exists at least one natural number P , for which $P!$ ends with K zeros, then how many distinct values can K assume?

Q12. DIRECTIONS *for question 12 to 17:* Select the correct alternative from the given choices.

Paykwik, a mobile wallet company offers a discount on movie tickets booked using its application. The company offers a 40% discount on the ticket price or a ₹150 discount, whichever is less on each ticket purchased through its application. On a certain day, if Virat purchased three tickets worth ₹330, ₹370 and ₹430 through the application, what is the approximate overall percentage of discount that Virat would get on the three transactions combined?

☐ a)
37.85%

☐ b)
37.95%

☐ c)
38.15%

☒ d)
38.05%

Q13. DIRECTIONS *for question 12 to 17:* Select the correct alternative from the given choices.

The ratio of the earnings of A, B and C is 2 : 3 : 5. If A increases his earnings by a certain amount, B by double that amount and C by triple that amount, which of the following could be the ratio of the earnings of A and C after the increase?

☐ a)

7 : 17

☐ b)

9 : 22

☐ c)

13 : 40

☒ d)

16 : 45

Q14. DIRECTIONS *for question 12 to 17:* Select the correct alternative from the given choices.

A trader claims to sell vegetables at a profit of 20% over his cost price. If he uses a false weight and always weighs only 800 gm instead of 1000 gm, what is his overall profit percentage?

- ☐ a) 25%
- ☐ b) 40%
- ☒ c) 50% ✓ **Your answer is correct**
- ☐ d) 60%

Q15. DIRECTIONS *for question 12 to 17:* Select the correct alternative from the given choices.

A rectangular sheet of paper was folded along the line joining the midpoints of its longer sides. The rectangle which resulted had the same ratio of its longer and shorter sides as that of the original rectangle. If the breadth of the original rectangle is $\sqrt{2}$ cm, find the area (in sq.cm) of the smaller rectangle.

☐ a) $\frac{3\sqrt{2}}{4}$

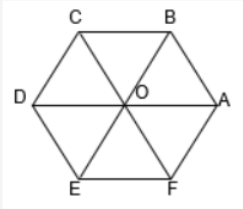
☐ b) $\sqrt{2}$

☐ c) $\frac{3\sqrt{2}}{2}$

☐ d) $4\sqrt{2}$

Q16. DIRECTIONS for question 12 to 17: Select the correct alternative from the given choices.

The figure below shows a network of roads in the form of a regular hexagon with 6 radial roads. Vinod starts at A and continually travels along ABCDEFA, while Raghav starts simultaneously at O and continually travels along OABOCDOEFO. If, by the time Vinod crosses C for the second time, Raghav overtakes Vinod on each of the roads AB, CD and EF, then which of the following could be the ratio of Raghav's speed to Vinod's speed?



- ☐ a) 2 : 1
- ☐ b) 3 : 2
- ☐ c) 4 : 3
- ☐ d) 5 : 3

Q17. DIRECTIONS *for question 12 to 17:* Select the correct alternative from the given choices.

In a textile shop, the number of shirts having a design is thrice the number of shirts not having a design. The shop also has sarees of exactly three colours – Brown, Black and Yellow. The number of brown sarees is half the number of black sarees and one-fifth of the number of yellow sarees. Find the number of shirts not having a design, given that the total number of shirts and sarees in the shop is 72 and the number of sarees that are handspun is four times that of those that are not handspun.

☐ a) **9**

☐ b) **8**

☐ c) **7**

☐ d) **6**

Q18.

DIRECTIONS *for question 18:* Select the correct alternative from the given choices.

There are three varieties of tea – I, II, III – available in the market. Mani bought a certain quantity of each variety. The average cost per kg of varieties I and II for the quantities he bought worked out to be ₹50. The same for varieties II and III is ₹60 and that for varieties I and III is ₹68. Which of the following cannot be the average cost per kg of all the three varieties of tea that he bought?

☐ a)

₹56

☐ b)

₹61

☐ c)

₹63

☐ d)

₹65

Q19.

DIRECTIONSfor question 19: Type in your answer in the input box provided below the question.

Usually, Mahesh takes 150 minutes to drive from his house to his in-laws' house. Last Sunday, after covering 20% of the distance to his in-laws' house, Mahesh's car developed an engine problem, which reduced his speed (when compared to his usual speed) by 15 km/hr for the remaining part of the journey. If it took Mahesh a total of 190 minutes to cover the entire distance, find the distance (in km) between his house and his in-laws' house.

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.

Pradeep wants to take a loan of ₹10,00,000 to buy a house. He approached two banks, A and B, for the loan. Both banks agreed to give the loan but at different terms. Bank A charges interest at 20% per annum, compounded annually, while bank B charges simple interest at 25% per annum. If Pradeep wants to repay the loan in a single payment at the end of n years, find the maximum value of n for which Bank A is a better preference for Pradeep when compared to Bank B.

- ☐ a) 2
- ☐ b) 3
- ☐ c) 4
- ☐ d) 5

Q21. DIRECTIONS *for questions 21 and 22:* Type in your answer in the input box provided below the question.

Find the sum of the first 20 terms of an arithmetic progression, if its fifth term is 11 and its 16th term is 39.

Q22. DIRECTIONS *for questions 21 and 22:* Type in your answer in the input box provided below the question.

The remainder obtained when 12345678987654321 is divided by 1001 is